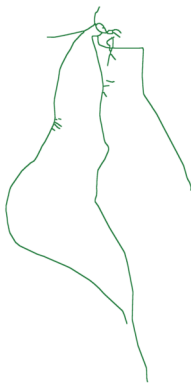


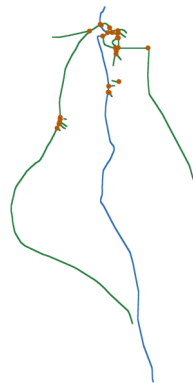
a Closed mask



b Pruned skeleton



c Descriptor regions



01 Close gaps with a 5 px elliptical kernel.

02 Skeletonize; prune ends for 15 passes.

03 Group pixels with ≥ 3 neighbors into regions.

04 Re-estimate the dominant path on this skeleton.

05 Retain off-path segments with ≥ 10 pixels.

06 Use valid local vectors for angles; else NaN.